

Priyanka Yadav

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Contact Information

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Citizenship

Indian

Personal

- Born on August 01, 2000.
- Known languages are English and Hindi.

Area of Research

- Multiscale modelling of Bio-nano systems

Systems of Interest

- Ion channels, solid-state nanopores, G-quadruplexes, DNA nanostructures, Proteins, Lipid bilayer membranes, drug-biomolecule interactions

Current Position

Senior Research Fellow at Department of Biotechnology, Indian Institute of Technology (IIT), Hyderabad, India, since July 2022.

- Advisor: Dr. Himanshu Joshi
- Thesis Title: Deciphering the structure and function of bio-nano systems using multiscale molecular simulation
- Area of study: Multiscale molecular simulations of biomolecular and bio-nano systems, focusing on the structure, dynamics, mechanics, and interactions of nucleic acids, membrane proteins, and nanopore systems.

Education

- M.Sc. (Life Science), 2021
[National Institute of Technology, Rourkela, India](#)
- B.Sc. (Botany Honours), 2019
[Maharani College, University of Rajasthan, Jaipur, Rajasthan, India](#)

Academic Honors

- Senior Research Fellowship awarded by CSIR in July, 2024.
- Qualified CSIR JRF, June 2021 (AIR 113)
- Qualified IIT-JAM in Biological Sciences, March 2019 (AIR 284)
- University Merit Position 5th in B.Sc. Honours Part 2nd

Publications

- Yadav P, Malik I, and Joshi H. “[Folding pathways and force-induced unfolding of neurodegeneration-associated GGGGCC microsatellite repeat RNA revealed by molecular simulations](#)”.
International Journal of Biological Macromolecules, 372 (2026), 153101.
- Gupta S, Yadav P, Joshi H, “[SketchDNA: A GUI-Enabled Toolkit for Multiscale Modeling of Topological DNA Structures](#)”, BioRxiv. 2026:2026-07.

Peer-Reviewed Publications

- Dahiya P, Yadav P, Singh A, Aggarwal S, Joshi H, Munde M. “[Protamine and Zinc Binding-Induced Conformational Effects on Insulin Analogs: An Integrated Experimental and Computational Study](#)”.
Physical Chemistry Chemical Physics, 2025, 27 (36), 19149–19162.
- D. Majhi, P. Yadav, Sahu PK, J Venkatesan, C Subrahmanyam. “[Chitosan-Based Nanoparticles in the Treatment of Alzheimer’s Disease](#)”.
Nanomedicine in the Treatment and Management of Alzheimer's Disease, pp. 137–151.
CRC Press, 2025.
- Prajapati A. Srivastava A, Sankaranarayanan A, Pebam M, Buddhiraju HS, Patra P, Yadav P, Joshi H, Rengan AK. “[High-Affinity DNA Nanomatrix: A Platform Technology for Synergistic Drug Delivery and Photothermal Therapy](#)”.
ACS Macro Letters, 2023 12 (2),255-262.

Manuscript Under Process

- D. Srivastava, P. Yadav, V. Nayak, V. Patil, A. Sharma, P. Mishra, H. Joshi, and A. Mishra, “Integrated in-silico and experimental analysis identifies bromocriptine and lifitegrast as OTUB1-targeting compounds in B-cell acute lymphoblastic leukemia”.
Manuscript under Review in Medical Oncology.
- Yadav P, Zehra M, and Joshi H. “Characterizing the conformational dynamics and ions permeation through Human NaV1.6 Sodium Channel”. *Manuscript in preparation.*
- Yadav P, Malik I and Joshi H. “Computational Investigation of Small-Molecule Binding to the C9orf72 GGGGCC RNA G-Quadruplex”. *Manuscript in preparation.*
- Yadav P and Joshi H. “Elucidate the molecular mechanism of electric field-driven ferritin capture and translocation through silicon nitride nanopores”. *Manuscript in preparation.*
- Yadav P, Tadanki AS, Modee R, H Joshi and Priyakumar D. “Machine Learning-Based Refinement of CGenFF Parameters for Enhanced Accuracy in Drug-Like Molecular Simulations”. *Manuscript in preparation.*

- Chakraborty S, Kannihalli A, Yadav P, Das S, K B Yahwan, Aggarwal S, Joshi H, and Ray S. “Circadian Regulation and Beta-Blocker Action: A Bidirectional Interplay Revealed by Metabolomics and Computational Modelling”. *Manuscript in preparation*.

Scientific Talks

- “Towards the fast and accurate force field parameters for small molecules using machine learning methods”. ML4Science conference, Trivandrum, Kerala, India. February, 2024.
- “Folding and unfolding of repeat RNA (G4C2) using MD simulations”. Horizons in Structural and Computational Biology, 2025, conference at IIT Hyderabad, India. March, 01, 2025.
- “Characterizing the folding and unfolding of G4C2 RNA repeats using a “computational microscope”. Institute of Protein Research, University of Osaka, Japan. December, 11, 2025.
- “Mechanistic Insights into the Ligand-Induced Unfolding of an RNA G-Quadruplex”. Inhouse Journal Club. Department of Biotechnology, IIT Hyderabad, India. March, 25, 2026.

Poster in Various Conference

- “MD@60: International conference on MD simulations” at JNC SAR, Bengaluru, India, from February 25-29, 2024.
- “IBS 2024: 46th Indian Biophysical Society (IBS) international conference” at University of Hyderabad and TIFR Hyderabad, India, from March 15-17, 2024.
- CSIR-Integrated skill program in “Emerging Topics in Molecular Dynamics Simulations” at CSIR National Chemical Laboratory, Pune, India, from January 20-24, 2025.
- “IBS 2025: 47th international conference” at IIT Madras, India, from March 6-9, 2025.
- “ML4Science conference 2024”, Thiruvananthapuram, India, February 01-04, 2024

Internships

NIT Rourkela, Odisha, India. July 2020 to December 2022

- Docking study of Tenofovir, Emtricitabine, Zidovudine, and Lamivudine against RNA-dependent RNA polymerase of SARS-Cov-2 and Reverse transcriptase of HIV-2, under the supervision of Dr. Suman Jha.

Nihon Nohyaku Co., Ltd. (NNC), Osaka, Japan. December 08-20, 2025

- Structure-based Drug Designing Strategies (SBDD) in the Development of Novel Drugs.

Software and Programming

- AMBER, NAMD, GROMACS, GAUSSIAN, VMD, AutoDock Tools
- Python, Unix Shell Scripting, Tcl, NAB, LaTeX

References

- Dr. Himanshu Joshi
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Website: [Molecular Simulation Lab](#)
- Dr. Indranil Malik
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Email: indranil@bt.iith.ac.in
Website: [The Malik Lab](#)
- Prof. U Deva Priyakumar
Center for Computational Natural Sciences,
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Hyderabad 500032, India
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